Kumar Gaurav

Backend Engineer — Distributed Systems & AI Agents

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github.com/D3S-Gaurav | d3s-gaurav.github.io | Codeforces: ATHEUS | LeetCode: ATHEUS_007

Available: 6-month internship from Dec 2026, full-time from Jul 2027

Education

Birla Institute of Technology, Mesra — Ranchi Aug 2023 – May 2027

B.Tech in Computer Science and Engineering · CGPA 7.88/10

Achievements

- Top 4% (rank **682 of 17,000+**) in Codeforces Round 1102 (Div. 2); 600+ DSA problems solved, 390+ on LeetCode.
- Finalist, SEBI Securities Market TechSprint 2026 (Global Fintech Fest): built RegOS Sentinel, a human-supervised agent system that turns SEBI circulars into cited, auditable compliance obligations.
- WorldQuant BRAIN Gold Level (8 peer-reviewed alphas).

Experience

Backend Engineer — SoilSense Mar 2026 – Sep 2026

Faculty-mentored AgriTech startup, BIT Mesra · Python · FastAPI · pandas · Gemini API

- Built the crop-suitability rules engine (FastAPI) across 22 crops on a public 2,200-sample dataset, using tolerance bands of 10% on NPK and 0.3 on pH to map each soil deficiency to an organic amendment.
- Combined 22 per-crop API calls into one batched endpoint on a pre-loaded dataset: **2.3x faster** (48 to 21 ms).
- Validated and type-cast the 5 sensor inputs before they reach the Gemini prompt, rejecting any unexpected fields.

Open Source — Oppia (Collaborator, merge access) · FOSSology Remote, Jan 2026 – Present

TypeScript · Angular · ESLint · CI

- Promoted to **Collaborator with merge access** on Oppia's Dev Workflow team (6.8k stars), after 3 merged PRs.
- Authored a custom ESLint rule that fails CI on undocumented `ViewEncapsulation.None`, a recurring source of CSS leaking across components; shipped with 10 tests and a 41-file legacy allowlist for incremental migration.
- Migrated FOSSology's Vagrant dev environment from end-of-life Ubuntu 20.04 to 22.04 to match its CI runners.

Research Intern — BIT Mesra May 2026 – Jul 2026

NLP / Content Moderation · Python · scikit-learn · LightGBM

- Built a comment-moderation classifier on 198k comments (TF-IDF plus 36 engineered signals); a LightGBM and Logistic Regression blend reached **0.836 macro-F1** and lifted the rare threat class from 0.23 to 0.65 F1.

Projects

Loom — Multi-Agent LLM Orchestration Platform Aug 2026 – Sep 2026

github.com/D3S-Gaurav/Loom · Go · TypeScript · Kafka · gRPC · Redis · PostgreSQL · Temporal

- Designed a planner, worker and validator agent loop: the planner turns one goal into a schema-validated task DAG, workers run independent nodes in parallel, and validators drive revisions, with Temporal crash recovery.
- Built an event pipeline from Redis through Kafka and a Go service to gRPC and WebSocket clients, with per-run sequence numbers for **idempotent replay**: a reconnecting client never applies an event twice.
- Wrote a non-blocking Go fan-out hub that evicts slow subscribers instead of stalling the Kafka consumer, verified under the race detector, and exported Prometheus metrics from the Go service.
- Exposed runs to AI clients such as Claude Code through a read-only, account-scoped MCP server.

ChatFlow — Real-Time Group Chat Jun 2026 – Jul 2026

github.com/D3S-Gaurav/chat-fullstack · TypeScript · Node.js · Socket.IO · Prisma · PostgreSQL · k6

- Load-tested with k6 to **5,000 concurrent sockets** and **~97k deliveries/s with zero message loss** on one Node process (p95 40 ms at 1k sockets); traced the ceiling to the single-threaded broadcast loop.
- Designed a typed Socket.IO event contract with Zod validation on every payload, JWT socket auth, scrypt hashing and tiered rate limits; covered by 32 integration tests in GitHub Actions.

Skills

Languages: Go, TypeScript, Python, SQL, C++ (competitive programming)

Backend: Node.js, FastAPI, gRPC, Kafka, Temporal, Redis, PostgreSQL, WebSockets, Socket.IO

AI: Multi-agent systems, MCP, structured outputs, LLM APIs (Gemini, OpenRouter)

Infra & Testing: Docker, GitHub Actions, Prometheus, k6, Linux